

# SAMA PATNAIK

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## EDUCATION

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### University of British Columbia

*Bachelor of Science in Statistics*

Sep 2022 – May 2027

Vancouver, BC

## EXPERIENCE

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### Data and Tool Developer Co-op

*Environment & Climate Change Canada*

May 2026 – August 2026

Vancouver, BC

- Supporting the development of forecast verification tools; Implemented FIFAFx Python GUI tool to automate data extraction and parsing, Excel report generation, and PowerPoint slide creation to support forecast communication for FIFA World Cup 2026.
- Developed Python ETL workflows to parse, extract and analyze 10 years of unstructured air quality bulletin text data; supporting decision making through PowerBI forecast verification dashboards.
- Cut manual errors by 40% and accelerated project delivery.

### Data and Survey Analyst Work Learn

*UBC Graduate School, Graduate + Postdoctoral Studies*

May 2025 – April 2026

Vancouver, BC

- Supporting the 2025 Graduate Career Outcomes Study and enterprise scale analytics on 20,000+ records.
- Automated Python ETL workflows integrating structured and unstructured data across 20,000+ records, using RESTful APIs for data retrieval; reduced manual workload by 25%
- Built interactive Tableau and R Shiny dashboards to visualize key trends in graduate employment and demographics.
- Designed and implemented validation routines and data health checks to ensure accuracy and traceability across reports.

### Environmental Data Scientist WLIURA

*Pelagic Ecosystems Lab, UBC Institute for the Oceans and Fisheries*

May 2024 – Aug 2024

Vancouver, BC

- Built an interactive R Shiny dashboard to visualize environmental data across 50+ regions for a team of 10+ researchers; authored data management documentation covering metric definitions, validation steps, and data logic to ensure reproducibility.

## PROJECTS

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### WiDS AirBnB Case Competition – UniLoft

*Python, XGBoost, Random Forest, Seaborn*

- Independently conducted market research and EDA on Airbnb listing data; developed and tested a stacked XGBoost and Random Forest model to address long-term housing needs; won Best Pitch Prize.

### Climate Data Warehouse

*Python, PostgreSQL, Docker, XGBoost, FastAPI, Streamlit*

- Designed and built a PostgreSQL star-schema data warehouse in Docker, ingesting NOAA GHCN-Daily station data for Canadian climate records. Built an XGBoost model to predict next-day temperature; exposed results via a FastAPI backend with a Streamlit interface for interactive exploration.

### Pricing Budapest – Airbnb Predictive Analysis

*R, Python, LASSO, OLS Regression, EDA, Jupyter Notebook*

- Formulated and tested hypotheses using LASSO and OLS regression models; conducted structured EDA and residual diagnostics to validate model assumptions and calibrate practical interpretation of results.

### WellChat – Mental Health Application

*Python, LangChain, Gemini Pro, OpenAI Embeddings, Pinecone, Flask, React*

- Built a health focused conversational AI using a RAG pipeline (document chunking, OpenAI embeddings, Pinecone retrieval) with multi-turn memory; tested outputs to verify Gemini Pro responses did not hallucinate beyond source data.

## TECHNICAL SKILLS

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**Languages:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), R, SQL, JavaScript

**Machine Learning:** XGBoost, Random Forest, LASSO/OLS Regression, C-V, feature engineering, EDA

**Dashboards & Visualization:** R Shiny, Tableau, Power BI, Plotly, Matplotlib, Seaborn

**Reproducibility & Tools:** Git, GitHub, Jupyter Notebook, RStudio, VS Code, data validation

**Certifications:** AWS Cloud Practitioner

**Awards:** Faculty of Science International Student Scholarship, WLIURA